

Problem Statement ID	26133
Title	Accessibility and quality of public healthcare services, particularly in rural and underserved areas
Organisation	Government of Maharashtra
Department	Maharashtra State Innovation Society, Dept. of Skills, Employment, Entrepreneurship & Innovation
Category / Theme	Software MedTech / BioTech / HealthTech

HOW TO USE THIS: In Gamma choose **Paste in text** → paste everything in PART 1 only → set **Cards = 6**, then Generate. PART 2 is your build guide (image prompts, diagrams, speaker notes) — do not paste it.

PART 1 — Paste this into Gamma (6 slides)

Keep the six headings exactly as written; SIH judges score against these fixed sections.

Slide 1 — Idea / Problem Statement

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Title: Accessibility and quality of public healthcare services, particularly in rural and underserved areas

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Team Name / Team ID: [FILL] / [FILL]

Idea title — HealthBridge: an offline-capable, ASHA-operated, ABDM/FHIR-linked patient record with explainable AI triage.

- **One-line pitch:** A longitudinal digital record that follows a rural patient across sub-centre → PHC → rural/district hospital, with danger-sign triage that reorders the queue and auto-escalates emergencies — strengthening the public system, not replacing it.

Slide 2 — Proposed Solution

What it is: Two connected surfaces on one backend — an offline-first field app for the ASHA/frontline worker, and a facility dashboard for doctors and admins.

How it solves the problem:

- **Longitudinal record keyed to ABHA number** — one patient, one thread, visible at every facility level. No repeated history-taking, no lost referrals, no fragmented records.
- **Explainable digital triage** — ASHA enters a free-text complaint in the local language; system flags Red / Yellow / Green *and shows why* (“Red: danger sign — unable to drink/feed”).
- **Severity-based queue** — triage reorders tokens automatically; Red cases jump the queue and fire an emergency escalation.
- **Assisted teleconsultation + referral tracking** — ASHA-mediated consults with store-and-forward fallback for low bandwidth; referrals tracked to completion.
- **Works without internet** — registration, triage and token creation run offline and sync when signal returns.
- **ABDM/FHIR interoperable** — records stored as FHIR resources, portable across the public health stack.

Innovation & uniqueness:

- **LLM extracts, rules decide.** The AI only pulls structured facts (symptoms, duration, age); a versioned IMNCI danger-sign rule table assigns severity. Safety is auditable, not a black box.
- **Offline triage engine** runs on-device — rural connectivity never disables the safety-critical path.
- **Adapter architecture** — ABDM, teleconsult and scheme checks sit behind swappable interfaces; today’s pilot becomes production with no rewrite.

Slide 3 — Technical Approach

Technology stack:

- **Frontend:** Next.js + TypeScript + Tailwind CSS; ASHA app ships as an installable **offline-first PWA**.
- **Backend:** FastAPI (Python) + REST APIs.
- **Databases:** PostgreSQL (FHIR-shaped records + queue/users) + Redis (queue/cache).
- **AI layer:** LangGraph orchestration, Whisper (local) for voice, local open-weight LLM (**Phi-4-mini via Ollama**) for symptom extraction — small footprint for modest PHC hardware, health-data stays on-premise. Cloud LLM only as optional online upgrade, never on the triage path.
- **Data model:** FHIR Patient / Encounter / Observation, everything keyed to ABHA number.

Methodology / process flow (triage):

Free-text complaint (local language) → **LLM** extracts structured facts only → **IMNCI rule table** assigns Red/Yellow/Green + records which rule fired → queue reorders; Red → auto-escalation → every decision written to an **audit log**. If the LLM is offline, ASHA fills a structured checklist and the *same* rules run.

Offline sync flow:

Local write queue (IndexedDB) → syncs on reconnect → append-only encounters, server-timestamp-wins conflict rule.

Slide 4 — Feasibility and Viability

Why it's feasible:

- Entirely open-source / free-tier stack; local LLM removes per-call API cost and keeps data on-premise.
- Runs on modest hardware already at PHCs; PWA installs on any Android phone the ASHA already owns.
- Built on approved national rails (ABDM/FHIR) — integration is a defined standard, not an invention.

Challenges & risks:

- Intermittent rural connectivity.
- Clinical trust in AI-assisted triage.
- ABDM sandbox onboarding is slow/bureaucratic.
- Low digital/health literacy among some users.

Strategies to overcome:

- **Offline-first by design** — core flows never require a live connection.
- **Explainable + audit-logged triage** (LLM-extracts-only + versioned rules) — defensible to a clinician.
- **Mock ABDM adapter from day one** — real gateway swaps in later without blocking deployment.
- **Local-language, voice-assisted, low-text UI** for the ASHA app.

Slide 5 — Impact and Benefits

Beneficiaries: Rural & underserved patients, ASHAs/frontline workers, PHC/CHC doctors, district health administrators.

Social impact:

- Faster care for emergencies — danger-sign cases surfaced and escalated in minutes, not missed.
- Continuity of care — history follows the patient, cutting repeat tests and diagnostic errors.
- Equitable access regardless of connectivity or language.

Health-system impact:

- Accountability & quality monitoring via a live 4-tile facility dashboard (triaged-by-severity, queue length, teleconsults, red cases escalated).
- Interoperable, ABDM-compliant data feeding the national health record.
- Reduced load on higher facilities through better triage and referral at source — shorter travel and waiting time.

Economic benefit:

- Lower out-of-pocket cost via fewer avoidable trips and duplicate tests; a scheme-status badge (PMJAY/state) flags entitlements at point of care. Near-zero marginal software cost per additional PHC.

Slide 6 — Research and References

Supporting rationale:

- Rural India faces persistent shortfalls in sub-centres, PHCs and specialist availability (Rural Health Statistics, MoHFW) — access and continuity, not just capacity, are the gap HealthBridge targets.
- IMNCI danger-sign protocols are the established, field-proven basis for frontline triage — the source for the rule table.
- ABDM/FHIR is the Government of India's approved standard for interoperable health records, making the longitudinal-record approach nationally aligned.

References:

- Ayushman Bharat Digital Mission (ABDM) — abdm.gov.in
- HL7 FHIR interoperability standard — hl7.org/fhir
- WHO / UNICEF IMNCI guidelines (Integrated Management of Neonatal & Childhood Illness)
- National Health Mission — ASHA programme, MoHFW — nhm.gov.in
- Rural Health Statistics, Ministry of Health & Family Welfare, Government of India

PART 2 — Build guide for the PPT (do not paste into Gamma)

A. Gamma generation settings

- **Import mode:** Paste in text → **Cards: 6** (one per slide). Turn OFF “auto-add cards” so it doesn’t split your slides.
- **Text amount:** “Preserve” or “Detailed” — not “Condense,” or it will cut your bullets.
- **Theme:** pick a clean medical/tech theme (e.g. Oasis, Chisel, or a blue/teal one). Set accent to a health blue/teal, not red.
- **Format:** 16:9. After generating, re-lock the six section headings to the exact SIH names — Gamma likes to rewrite them.
- **Font:** keep one sans-serif (Inter/Poppins). Body text ≥ 16pt so it reads from the back of a room.

B. The one slide that wins or loses it — Slide 3 diagram

Judges are technical. Turn the triage flow into a left-to-right diagram instead of bullets. Paste this as a diagram prompt in Gamma (or draw in draw.io / Excalidraw and insert):

“Horizontal flowchart, 5 boxes with arrows: (1) Free-text complaint in local language → (2) LLM extracts structured facts only [symptoms, duration, age] → (3) IMNCI rule table assigns Red/Yellow/Green + logs which rule fired → (4) Queue reorders; Red = auto-escalation → (5) Audit log. Add a dashed fallback arrow from box 1 to box 3 labelled ‘offline: manual checklist → same rules’. Colour box 3 output chips red/amber/green.”

C. Suggested image / visual per slide

Slide 1: Clean hero: ASHA worker with a phone in a rural setting, or a simple India-map + health-cross motif. Keep it light so the PS table reads.

Slide 2: Two-device mockup: phone (ASHA app) + laptop (facility dashboard) side by side, one arrow between them to a central record.

Slide 3: The flowchart above (primary). Optionally a small stack-diagram of the tech layers.

Slide 4: A simple 2-column ‘Risk → Mitigation’ table; Gamma renders these cleanly. No stock photo needed.

Slide 5: Icon row (patient / ASHA / doctor / admin) + the 4 dashboard tiles as stat cards. Numbers optional — don’t invent stats.

Slide 6: Plain reference list; add small logos of ABDM / NHM / WHO only if genuine. No decorative image.

D. 30-second speaker note per slide

Slide 1: “PS 26133 — rural healthcare access for Government of Maharashtra. Our answer is HealthBridge: one patient record that follows the person across every facility, with AI triage that catches emergencies. We strengthen the public system, we don’t replace it.”

Slide 2: “An ASHA opens the offline app, registers a patient against their ABHA number, and types the complaint in the local language. The system flags severity and tells her why. Red cases jump the queue and escalate. The doctor sees the full history on the dashboard.”

Slide 3: “The key safety choice: the LLM never decides severity — it only extracts facts. A versioned IMNCI rule table makes the call and logs which rule fired. That runs on-device, so it works with zero connectivity.”

Slide 4: “It’s buildable now — open-source stack, runs on hardware PHCs already have, on standards ABDM already defines. The real risks are connectivity, clinical trust and ABDM onboarding — and we’ve designed each one out: offline-first, explainable triage, mock adapter from day one.”

Slide 5: “Patients get faster, continuous, cheaper care. The system gets accountability through a live dashboard and interoperable records. And each new PHC costs almost nothing to add.”

Slide 6: “This isn’t speculative — it sits on MoHFW rural-health data, WHO IMNCI protocols and India’s own ABDM/FHIR standard.”

E. Before you submit — checklist

- Fill Team Name + Team ID on Slide 1.
- Confirm the six headings match the SIH template wording exactly.
- Add the YouTube demo link on Slide 1 or 6 if your team records one (the PS has a YouTube-link field).
- Export from Gamma as PDF **and** PPTX — SIH portals sometimes want one or the other.
- Keep it to exactly 6 slides. Do not add a thank-you slide.
- Do not invent statistics — every number must be sourced or left out.